UNIT-I
Introduction to IoT

IoT comprises things that have unique identities and are connected to internet. By 2020 there will be a total of 50 billion devices /things connected to internet. IoT is not limited to just connecting things to the internet but also allow things to communicate and exchange data.

1.1. Definition

A dynamic global network infrastructure with self-configuring capabilities based on standard and interoperable communication protocols where physical and virtual "things" have identities, physical attributes and virtual personalities and use intelligent interfaces, and are seamlessly integrated into information network, often communicate data associated with users and their environments.

1.1.1 Alternative Definition

The Internet of Things (IoT) is the network of physical objects that contain embedded technology to communicate and sense or interact with their internal states or the external environment. (or)
Connecting physical objects/ things over the internet in real time.

1.2. Characteristics of IoT

1) **Dynamic & Self Adapting**: IoT devices and systems may have the capability to dynamically adapt with the changing contexts and take actions based on their operating conditions, user 's context or sensed environment.
   Ex: Surveillance system – can adapt normal (day) and night time modes
2) **Self-Configuring:** allowing a large number of devices to work together to provide certain functionality.
   Ex: smart home lighting system. When introduce a new smart light bulb into the network, it doesn't require manual configuration. Instead, it automatically detects the presence of the smart home hub, connects to the network, obtains a unique IP address, and registers itself with the hub.

3) **Inter Operable Communication Protocols:** support a number of interoperable communication protocols and can communicate with other devices and also with infrastructure.
   Ex: Using an IoT gateway to translate between different smart home protocols, like Zigbee and Bluetooth, allowing devices from various brands to function together.
4) **Unique Identity:** Each IoT device has a unique identity and a unique identifier (IP address).
   Ex: Asset Tracking: Each item or piece of equipment in a logistics company is tagged with an RFID label carrying a unique code, so its location and status can be updated continually in a database.
5) **Integrated into Information Network:** that allow them to communicate and exchange data with other devices and systems.
   Ex: Smart Home Appliances: LG and Samsung smart appliances connect to user phones and cloud servers, reporting status, receiving commands, and interacting with digital assistants like Alexa.

1.3. Applications of IoT/ Domain Specific IoT

1) **Home Automation:**
   a) **Smart Lighting:** helps in saving energy by adapting the lighting to the ambient conditions and switching on/off or diming the light when needed.
   b) **Smart Appliances:** make the management easier and also provide status information to the users remotely.
   c) **Intrusion Detection:** use security cameras and sensors (PIR sensors and door sensors) to detect intrusion and raise alerts. Alerts can be in the form of SMS or email sent to the user.
   d) **Smoke/Gas Detectors:** Smoke detectors are installed in homes and buildings todetect smoke that is typically an early sign of fire. Alerts raised by smoke detectors can be in the form of signals to a fire alarm system. Gas detectors can detect the presence of harmful gases such as CO, LPG etc.,
2) **Cities:**
   a) **Smart Parking:** make the search for parking space easier and convenient for drivers.

Smart parking are powered by IoT systems that detect the no. of empty parking slots and send information over internet to smart application backends.

b) **Smart Lighting:** for roads, parks and buildings can help in saving energy.

c) **Smart Roads:** Equipped with sensors can provide information on driving condition, travel time estimating and alert in case of poor driving conditions, traffic condition and accidents.

d) **Structural Health Monitoring:** uses a network of sensors to monitor the vibration levels in the structures such as bridges and buildings.

e) **Surveillance:** The video feeds from surveillance cameras can be aggregated in cloud based scalable storage solution.

f) **Emergency Response:** IoT systems for fire detection, gas and water leakage detection can help in generating alerts and minimizing their effects on the critical infrastructures.

3) Environment:

a) **Weather Monitoring:** Systems collect data from a no. of sensors attached and send the data to cloud-based applications and storage back ends. The data collected in cloud can then be analyzed and visualized by cloud-based applications.

b) **Air Pollution Monitoring:** System can monitor emission of harmful gases (CO2, CO, NO, NO2 etc.,) by factories and automobiles using gaseous and meteorological sensors. The collected data can be analyzed to make informed decisions on pollutions control approaches.

c) **Noise Pollution Monitoring:** Due to growing urban development, noise levels in cities have increased and even become alarmingly high in some cities. IoT based noise pollution monitoring systems use a no. of noise monitoring systems that are deployed at different places in a city. The data on noise levels from the station is collected on servers or in the cloud. The collected data is then aggregated to generate noise maps.

d) **Forest Fire Detection:** Forest fire can cause damage to natural resources, property and human life. Early detection of forest fire can help in minimizing damage.

e) **River Flood Detection:** River floods can cause damage to natural and human resources and human life. Early warnings of floods can be given by monitoring the water level and flow rate. IoT based river flood monitoring system uses a no. ofsensor nodes that

monitor the water level and flow rate sensors.

4) **Energy:**
   a) **Smart Grids:** is a data communication network integrated with the electrical grids that collects and analyze data captured in near-real-time about power transmission, distribution and consumption. Smart grid technology provides predictive information and recommendations to utilities, their suppliers, and their customers on how best to manage power. By using IoT based sensing and measurement technologies, the health of equipment and integrity of the grid can be evaluated.
   b) **Renewable Energy Systems:** IoT based systems integrated with the transformers at the point of interconnection measure the electrical variables and how much power is fed into the grid. For wind energy systems, closed-loop controls can be used to regulate the voltage at point of interconnection which coordinate wind turbine outputsand provides power support.
   c) **Prognostics:** In systems such as power grids, real-time information is collected using specialized electrical sensors called Phasor Measurement Units (PMUs) at thesubstations. The information received from PMUs must be monitored in real- time for estimating the state of the system and for predicting failures.

5) **Retail:**
   a) **Inventory Management:** IoT systems enable remote monitoring of inventory using data collected by RFID readers.
   b) **Smart Payments:** Solutions such as contact-less payments powered by technologiessuch as Near Field Communication (NFC) and Bluetooth.
   c) **Smart Vending Machines:** Sensors in a smart vending machine monitors its operations and send the data to cloud which can be used for predictive maintenance.

6) **Logistics:**
   a) **Route generation & scheduling:** IoT based system backed by cloud can provide firstresponse to the route generation queries and can be scaled up to serve a large transportation network.

b) **Fleet Tracking:** Use GPS to track locations of vehicles in real-time.

c) **Shipment Monitoring:** IoT based shipment monitoring systems use sensors such as temp, humidity, to monitor the conditions and send data to cloud, where it can be analyzed to detect food spoilage.

d) **Remote Vehicle Diagnostics:** Systems use on-board IoT devices for collecting data on Vehicle operations (speed, RPM etc.,) and status of various vehicle subsystems.

7) **Agriculture:**

a) **Smart Irrigation:** to determine moisture amount in soil.

b) **Green House Control:** to improve productivity.

8) **Industry:**

a) Machine diagnosis and prognosis

b) Indoor Air Quality Monitoring

9) **Health and Lifestyle:**

a) Health & Fitness Monitoring

b) Wearable Electronics

1.4. Physical Design Of IoT

1.4.1. Things in IoT:

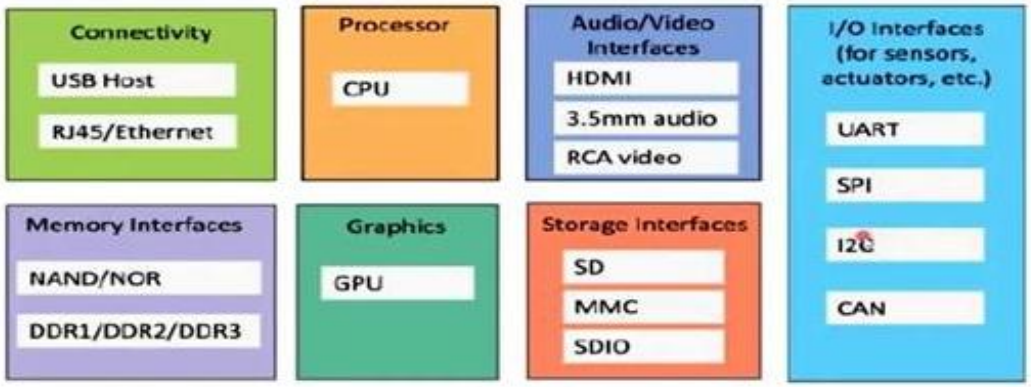


Figure 1.1: Generic Block Diagram of IoT

The things in IoT refers to IoT devices which have unique identities and perform remote sensing, actuating and monitoring capabilities. IoT devices can exchange data with other connected devices applications. It collects data from other devices and process data either locally or remotely.

An IoT device may consist of several interfaces for communication to other devices both wired and wireless.

These includes (i) I/O interfaces for sensors, (ii) Interfaces for internet connectivity

(iii) memory and storage interfaces and (iv) audio/video interfaces.

1.4.2. IoT Protocols

a) **Link Layer:** Protocols determine how data is physically sent over the network‘s physical layer or medium. Local network connects to which host is attached. Hosts on the same link exchange data packets over the link layer using link layer protocols. Link layer determines how packets are coded and signaled by the h/w device over the medium to which the host is attached.

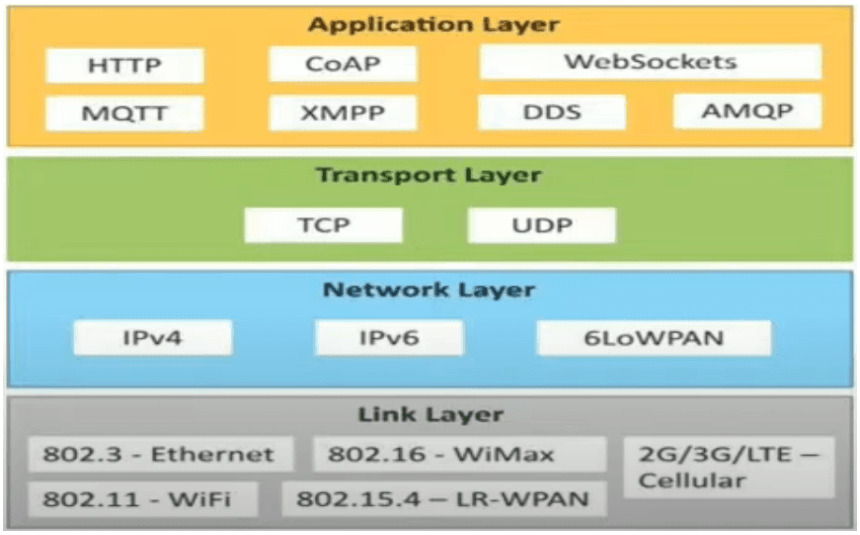


Figure 1.2. IoT Protocols

a) Link Layer Protocols:

- 802.3-Ethernet: IEEE802.3 is collection of wired Ethernet standards for the link layer. Eg: 802.3 uses co-axial cable; 802.3i uses copper twisted pair connection; 802.3j uses fiber optic connection; 802.3ae uses Ethernet over fiber.
- 802.11-WiFi: IEEE802.11 is a collection of wireless LAN(WLAN) communication standards including extensive description of link layer. Eg: 802.11a operates in 5GHz band, 802.11b and 802.11g operates in 2.4GHz band, 802.11n operates in 2.4/5GHz band, 802.11ac operates in 5GHz band, 802.11ad operates in 60Ghzband.
- 802.16 - WiMax: IEEE802.16 is a collection of wireless broadband standards including exclusive description of link layer. WiMax provide data rates from 1.5 Mb/s to 1Gb/s.
- 802.15.4-LR-WPAN: IEEE802.15.4 is a collection of standards for low-rate wireless personal area network (LR-WPAN). Basis for high level communication protocols such asZigBee. Provides data rate from 40kb/s to250kb/s.
- 2G/3G/4G-Mobile Communication: Data rates from 9.6kb/s(2G) to up to100Mb/s(4G).

b) Network/Internet Layer: Responsible for sending IP datagrams from source n/w to destination n/w. Performs the host addressing and packet routing. Datagrams contains source and destination address.

Protocols:

i. **IPv4:** Internet Protocol version4 is used to identify the devices on a n/w using a

hierarchical addressing scheme. 32-bit address. Allows total of 2^{32}addresses.

ii. **IPv6:** Internet Protocol version6 uses 128-bit address scheme and allows 2^{128}addresses.

iii. **6LOWPAN:**(IPv6over Low power Wireless Personal Area Network) operates in 2.4 GHz frequency range and data transfer 250 kb/s.

c) Transport Layer: Provides end-to-end message transfer capability independent of the underlying n/w. Set up on connection with ACK as in TCP and without ACK as in UDP. Provides functions such as error control, segmentation, flow control and congestion control.

Protocols:

i. **TCP:** Transmission Control Protocol used by web browsers (along with HTTP and HTTPS), email (along with SMTP, FTP). Connection oriented and stateless protocol. IP Protocol deals with sending packets, TCP ensures reliable transmission of protocols in order. Avoids n/w congestion and congestion collapse.

ii. **UDP:** User Datagram Protocol is connectionless protocol. Useful in time sensitive applications, very small data units to exchange. Transaction oriented and stateless protocol. Does not provide guaranteed delivery.

d) Application Layer: Defines how the applications interface with lower layer protocols to send data over the n/w. Enables process-to-process communication using ports.

Protocols:

i. **HTTP:** Hyper Text Transfer Protocol that forms foundation of WWW. Follow request-response model Stateless protocol.

ii. **CoAP:** Constrained Application Protocol for machine-to- machine(M2M) applications with constrained devices, constrained environment and constrained n/w. Uses client-server architecture.

iii. **WebSocket:** allows full duplex communication over a single socket connection.

iv. **MQTT:** Message Queue Telemetry Transport is light weight messaging protocol based on publish-subscribe model. Uses client server architecture. Well suited for constrained environment.

v. **XMPP:** Extensible Message and Presence Protocol for real time communication and streaming XML data between network entities. Support client-server and server-server communication.

vi. **DDS:** Data Distribution Service is data centric middleware standards for device-to-

device or machine-to-machine communication. Uses publish-subscribe model.

vii. **AMQP:** Advanced Message Queuing Protocol is open application layer protocol for business messaging. Supports both point-to-point and publish-subscribe model.

1.5. Logical Design of IoT

Refers to an abstract represent of entities and processes without going into the low level specifies of implementation.

1) IoT Functional Blocks 2) IoT Communication Models 3) IoT Comm. APIs

1.5.1. IoT Functional Blocks:

Provide the system the capabilities for identification, sensing, actuation, communication and management.

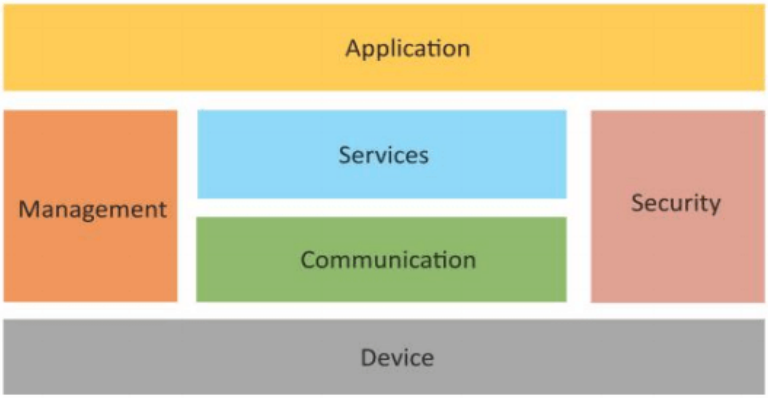


Figure 1.3: IoT Functional Blocks

- **Device:** An IoT system comprises of devices that provide sensing, actuation, monitoring and control functions.
- **Communication:** handles the communication for IoT system.
- **Services:** for device monitoring, device control services, data publishing services and services for device discovery.
- **Management:** Provides various functions to govern the IoT system.
- **Security:** Secures IoT system and priority functions such as authentication, authorization, message and context integrity and data security.
- **Application:** IoT application provide an interface that the users can use to control and

monitor various aspects of IoT system.

1.5.2. IoT Communication Models

a) Request-Response Model:

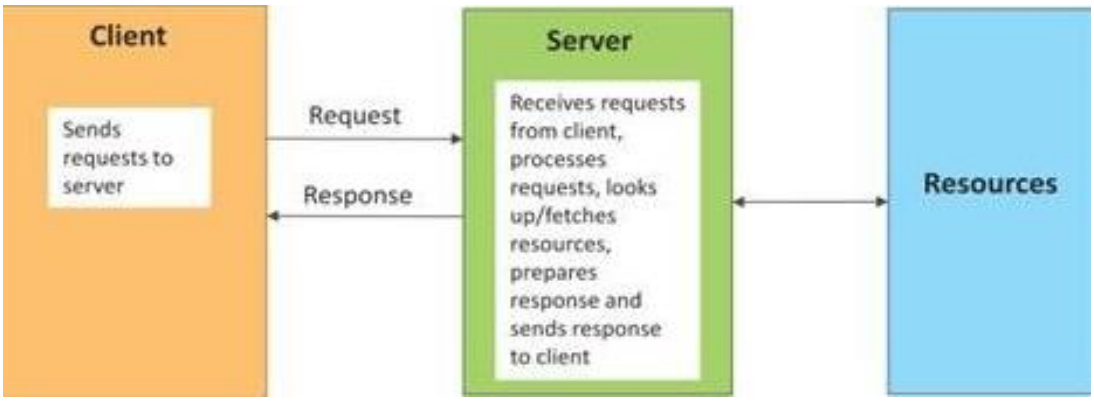


Figure 1.4: Request-response model

In which the client sends request to the server and the server replies to requests. Is a stateless communication model and each request-response pair is independent of others.

b) Publish-Subscribe Model:

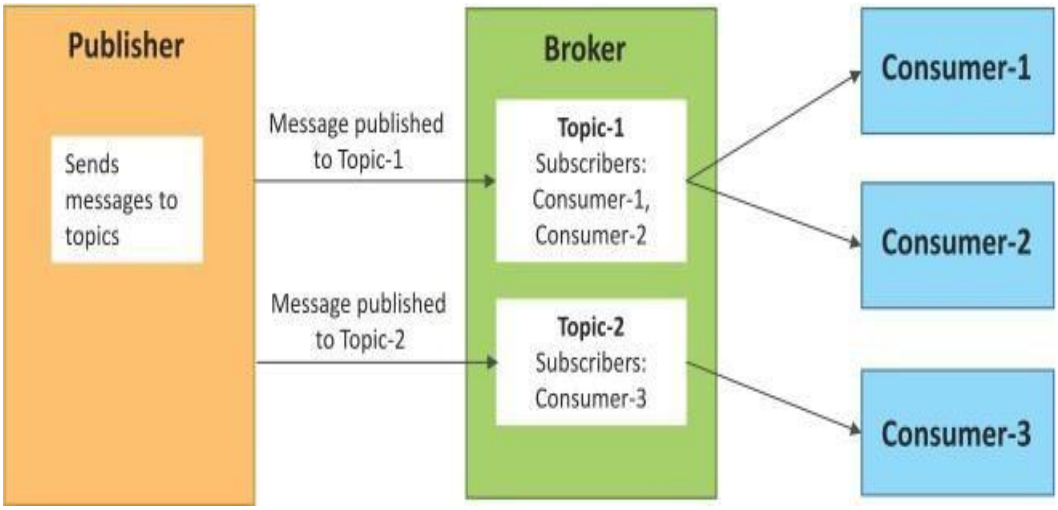


Figure 1.5: Publish-Subscribe Model

Involves publishers, brokers and consumers. Publishers are source of data. Publishers send data to the topics which are managed by the broker. Publishers are not aware of the

consumers. Consumers subscribe to the topics which are managed by the broker. When the broker receives data for a topic from the publisher, it sends the data to all the subscribed consumers.

c) Push-Pull Model:

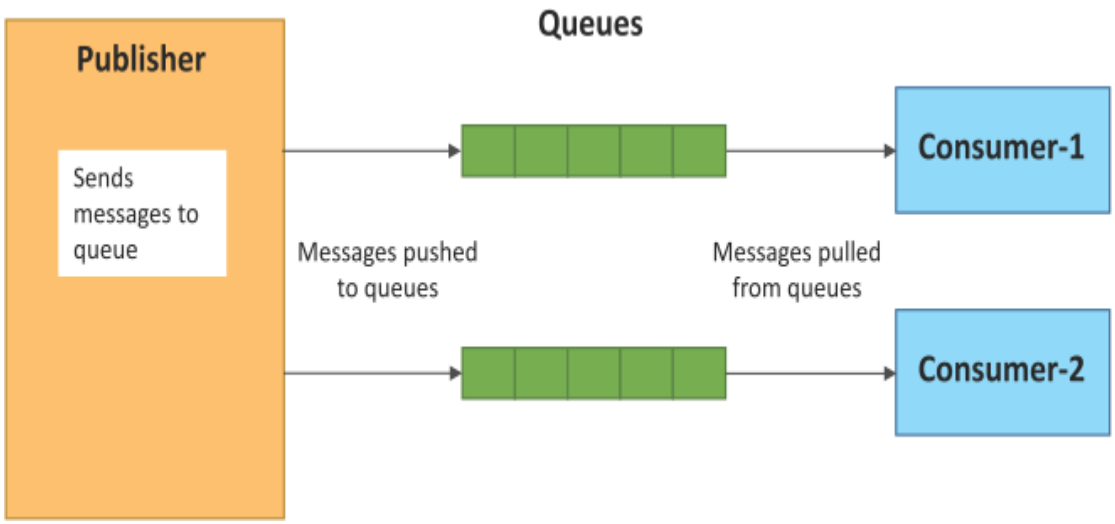


Figure 1.6: Push-pull model

In which data producers push data to queues and consumers pull data from the queues. Producers do not need to aware of the consumers. Queues help in decoupling the message between the producers and consumers.

d) Exclusive Pair:



Figure 1.7: Exclusive Pair Model

Is bi-directional, fully duplex communication model that uses a persistent connection between the client and server. Once connection is set up it remains open until the client

send a request to close the connection. Is a stateful communication model and server is aware of all the open connections.

1.5.3.IoT Communication APIs

a) REST based communication APIs (Request-Response Based Model)

Representational State Transfer (REST) is a set of architectural principles by which we can design web services and web APIs that focus on a system 's resources and have resource states are addressed and transferred.

The REST architectural constraints: Fig. shows communication between client server with REST APIs.

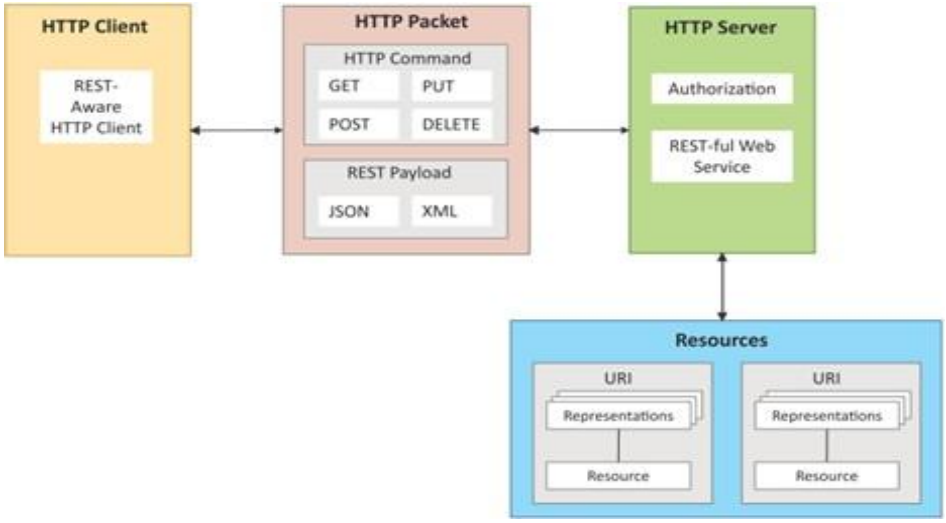


Figure: 1.8: REST based Communication API

Client-Server: The principle behind client-server constraint is the separation of concerns. Separation allows client and server to be independently developed and updated.

Stateless: Each request from client to server must contain all the info. Necessary to understand the request, and cannot take advantage of any stored context on the server.

Cache-able: Cache constraint requires that the data within a response to a request be implicitly or explicitly labeled as cache-able or non-cacheable. If a response is cache-able, then a client cache is given the right to reuse that response data for later, equivalent requests. **Layered System:** constraints the behavior of components such that each component cannot see beyond the immediate layer with which they are interacting.

User Interface: constraint requires that the method of communication between a client and a

server must be uniform.

Code on Demand: Servers can provide executable code or scripts for clients to execute in their context. This constraint is the only one that is optional.

Request-Response model used by REST:

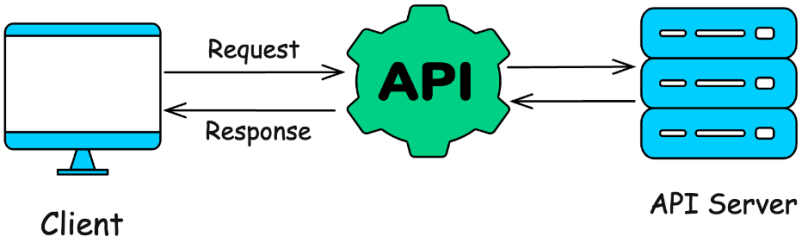


Figure 1.9: Request-Response Model used by REST

RESTful webservice is a collection of resources which are represented by URIs. RESTful web API has a base URI (e.g: http://example.com/api/tasks/). The clients and requests to these URIs using the methods defined by the HTTP protocol (e.g: GET, PUT, POST or DELETE). A RESTful web service can support various internet media types.

b) WebSocket Based Communication APIs: Web Socket APIs allow bi- directional, full duplex communication between clients and servers. Web Socket APIs follow the exclusive pair communication model.

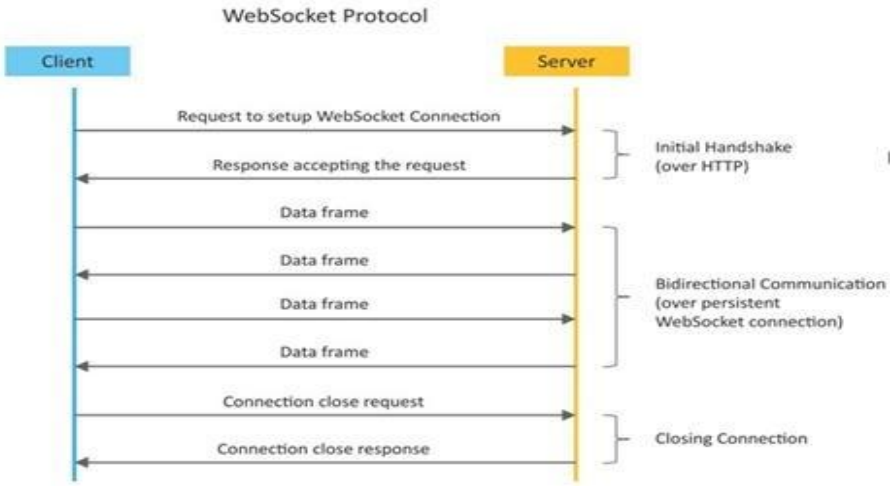


Figure 1.9: Web socket-based Communication API

1.6. IoT Enabling Technologies

IoT is enabled by several technologies including Wireless Sensor Networks, Cloud Computing, Big Data Analytics, Embedded Systems, Security Protocols and architectures, Communication Protocols, Web Services, Mobile internet and semantic search engines.

1) **Wireless Sensor Network (WSN):** Comprises of distributed devices with sensors which are used to monitor the environmental and physical conditions. Zig Bee is one of the most popular wireless technologies used by WSNs.

   WSNs used in IoT systems are described as follows:

   - Weather Monitoring System: in which nodes collect temp, humidity and other data, which is aggregated and analyzed.
   - Indoor air quality monitoring systems: to collect data on the indoor air quality and concentration of various gases.
   - Soil Moisture Monitoring Systems: to monitor soil moisture at various locations.
   - Surveillance Systems: use WSNs for collecting surveillance data (motion data detection).
   - Smart Grids: use WSNs for monitoring grids at various points.
   - Structural Health Monitoring Systems: Use WSNs to monitor the health of structures (building, bridges) by collecting vibrations from sensor nodes deployed at various points in the structure.

2) **Cloud Computing:** Services are offered to users in different forms.

   - Infrastructure-as-a-service (IaaS): provides users the ability to provision computing and storage resources. These resources are provided to the users as a virtual machine instances and virtual storage.
   - Platform-as-a-Service (PaaS): provides users the ability to develop and deploy application in cloud using the development tools, APIs, software libraries and services provided by the cloud service provider.
   - Software-as-a-Service (SaaS): provides the user a complete software application or the user interface to the application itself.

3) **Big Data Analytics:** Some examples of big data generated by IoT are
    - Sensor data generated by IoT systems.
    - Machine sensor data collected from sensors established in industrial and energy systems.
    - Health and fitness data generated IoT devices.
    - Data generated by IoT systems for location and tracking vehicles.
    - Data generated by retail inventory monitoring systems.

4) **Communication Protocols:** form the back-bone of IoT systems and enable network connectivity and coupling to applications.
    - Allow devices to exchange data over network.
    - Define the exchange formats, data encoding addressing schemes for device and routing of packets from source to destination.
    - It includes sequence control, flow control and retransmission of lost packets.

5) **Embedded Systems:** is a computer system that has computer hardware and software embedded to perform specific tasks. Embedded System range from low-cost miniaturized devices such as digital watches to devices such as digital cameras, POS terminals, vending machines, appliances etc.,

1.7. IoT Levels and Deployment Templates

1) **IoT Level1:** System has a single node that performs sensing and/or actuation, stores data, performs analysis and host the application as shown in fig. Suitable for modeling low cost and low complexity solutions where the data involved is not big and analysis requirement are not computationally intensive. An e.g., of IoT Level1 is home automation.

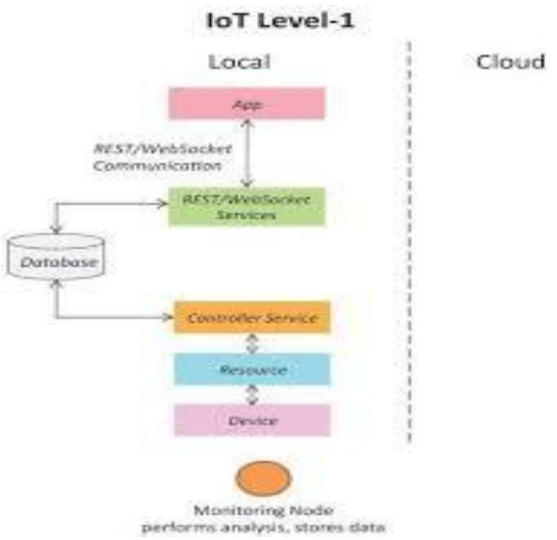


Figure 1.10: IoT deployment level-1-Template

2) **IoT Level2:** has a single node that performs sensing and/or actuating and local analysis as shown in fig. Data is stored in cloud and application is usually cloud based. Level2 IoT systems are suitable for solutions where data are involved is big, however, the primary analysis requirement is not computationally intensive and can be done locally itself. An e,g.**,** of Level2 IoT system for Smart Irrigation.

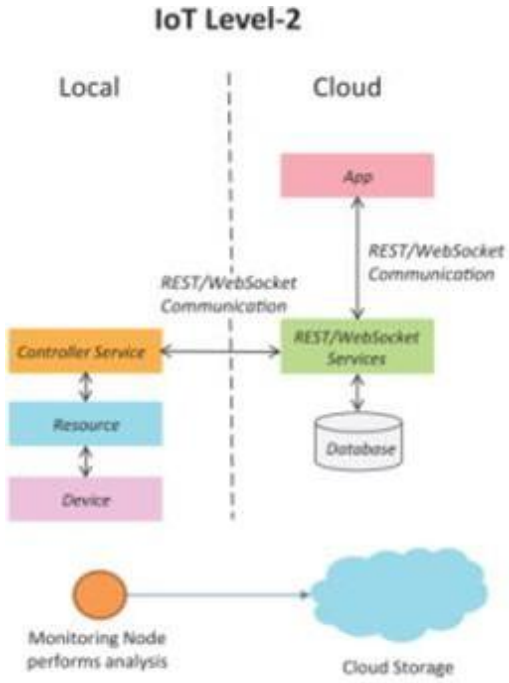


Figure 1.11: IoT deployment level-2-Template

3) **IoT Level3:** system has a single node. Data is stored and analyzed in the cloud application is cloud based as shown in fig. Level3 IoT systems are suitable for solutions where the data involved is big and analysis requirements are computationally intensive. An example of IoT level3 system for tracking package handling.

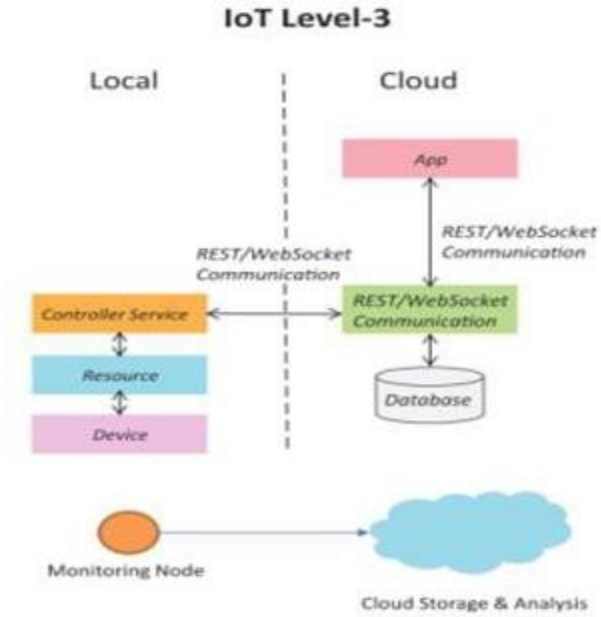


Figure 1.12: IoT deployment level-3-Template

4) **IoT Level4:** System has multiple nodes that perform local analysis. Data is stored in the cloud and application is cloud based as shown in fig. Level4 contains local and cloud-based observer nodes which can subscribe to and receive information collected in the cloud from IoT devices. An example of a Level4 IoT system for Noise Monitoring.

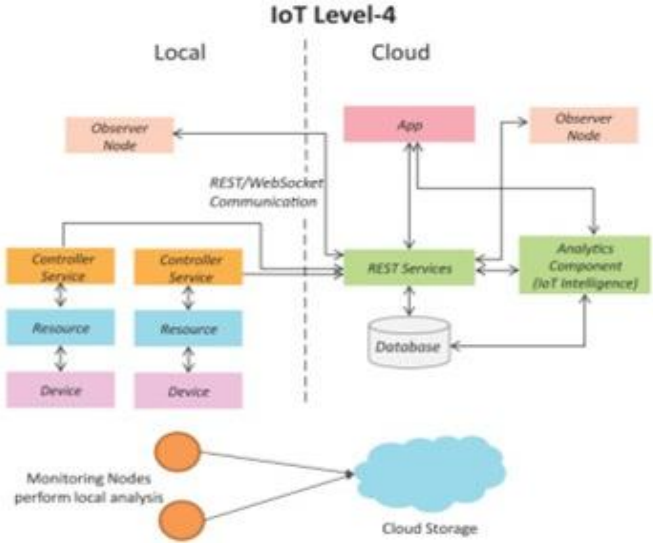


Figure 1.13: IoT deployment level-4-Template

5) **IoT Level5:** System has multiple end nodes and one coordinator node as shown in fig. The end nodes that perform sensing and/or actuation. Coordinator node collects data from the end nodes and sends to the cloud. Data i s stored and analyzed in the cloud and application is cloud based. Level5 IoT systems are suitable for solution based on wireless sensor network, in which data involved is big and analysis requirements are computationally intensive. An example of Level5 system for Forest Fire Detection.

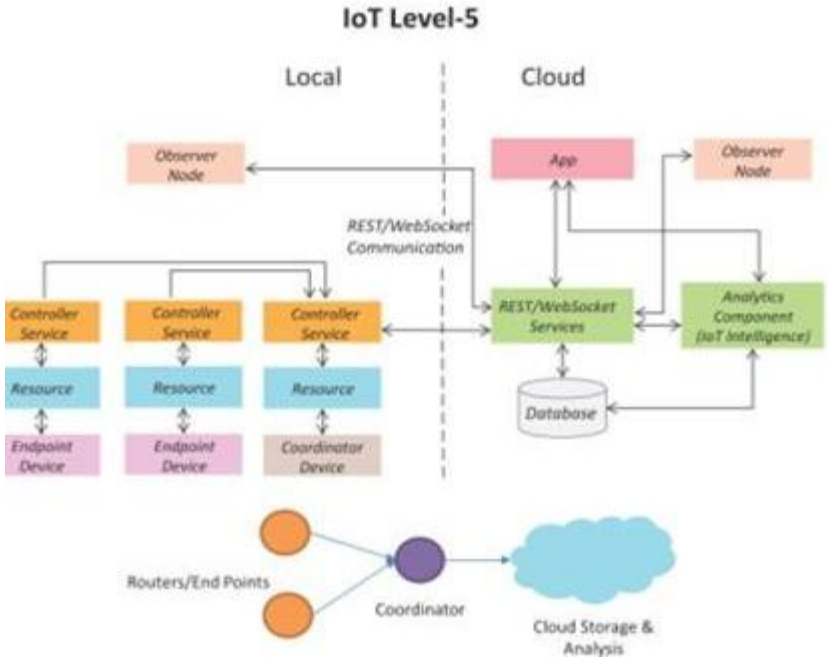


Figure 1.14: IoT deployment level-5-Template

6) **IoT Level6:** System has multiple independent end nodes that perform sensing and/or actuation and sensed data to the cloud. Data is stored in the cloud and application is cloud based as shown in fig. The analytics component analyses the data and stores the result in the cloud data base. The results are visualized with cloud-based application. The centralized controller is aware of the status of all the end nodes and sends control commands to nodes. An example of a Level6 IoT system for Weather Monitoring System.

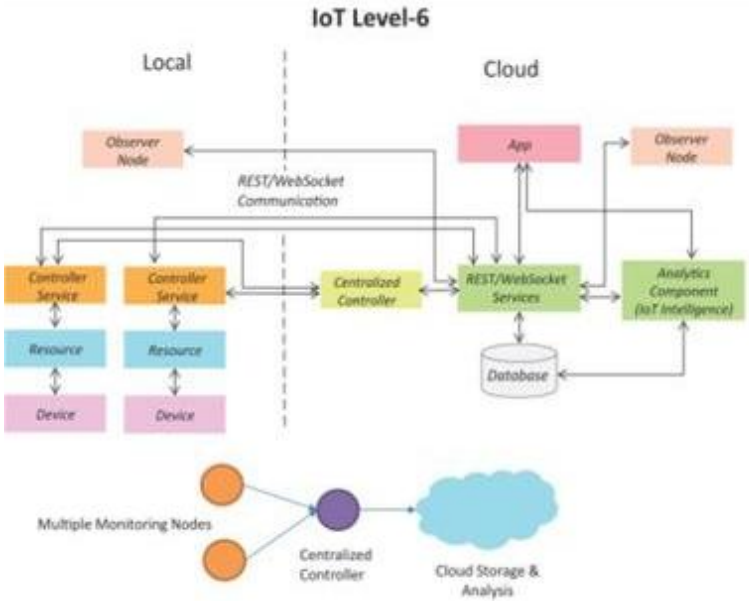


Figure 1.15: IoT deployment level-6-Template

1.8. IoT stack

7 layers of IoT protocol stack as shown in the figure below.

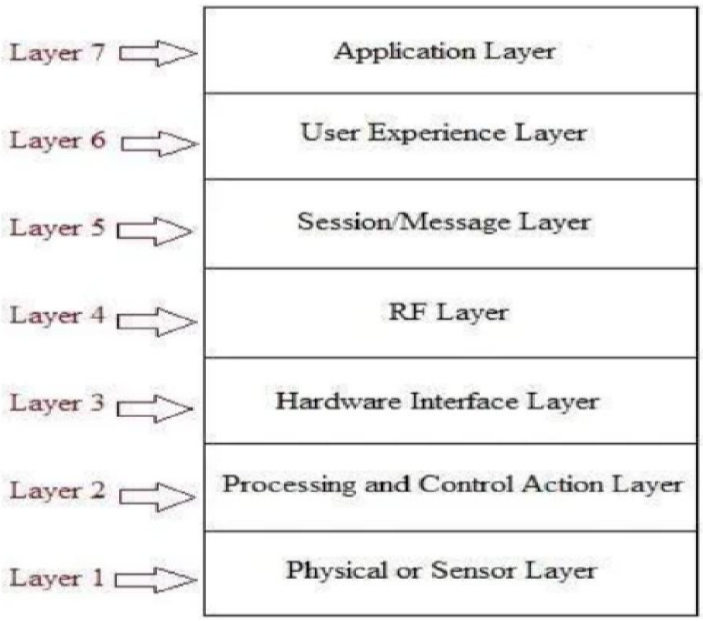


Figure 1.16: 7 Layer IoT stack

a) IoT Stack Layer 1: physical layer:

Physical or sensor Similar to OSI physical layer, this IoT layer 1 interfaces with physical components. The physical components are mainly sensors such as humidity sensor, temperature sensor, pressure sensor, heartrate sensor, pH sensor, odour sensor etc. The sensors are used for sensing of various parameters as per application of use. There are plenty of sensors available for same functionality and hence appropriate selection of sensor is done based on cost and quality. It is this layer-1 which provides sensed data to IoT stack for further processing.

b) IoT Stack Layer 2: Processing and control layer:

The data provided by layer-1 using sensors are processed at this layer. Microcontroller/Processor and operating system play vital role at this layer. Various development kits can be used for this purpose such as Arduino, Node MCU (based on ESP32 or ESP8266), ARM, PIC etc. Typical operating systems used are Android, Linux, IOS etc.

c) IoT Stack Layer 3: Hardware Interface layer:

This layer includes components or interfaces used for communication such as RS232, RS485, SPI, I2C, CAN, SCI etc. These interfaces are used for serial or parallel communication at various baud rates in synchronous/asynchronous modes. The above-mentioned interface protocols ensure flawless communication.

d) IoT Stack Layer 4: RF layer:

This radio frequency layer houses RF technologies based on short range or long range and data rate desired by the application of use. The common indoor RF/wireless technologies include WIFI, Bluetooth, Zigbee, Zwave, NFC, RFID etc. The common outdoor RF cellular technologies include GSM/GPRS, CDMA, LTE-M, NB-IoT, 5G etc. RF layer does communication of data using radio frequency-based EM waves. There is another technology which uses light waves for data communication. This light-based data communication is referred as LiFi.

e) IoT Stack Layer 5: Session/Message Layer:

This layer deals with various messaging protocols such as MQTT, CoAP, HTTP, FTP (or Secured FTP), SSH etc. It defines how messages are broadcasted to the cloud. Refer architectures of MQTT

protocol and CoAP protocol.

f) IoT Stack Layer 6: User experience layer:

This layer deals with providing best experience to the end users of IoT products. To fulfill this, this layer takes care of rich UI designs with lots of features. Various languages and tools are developed for the design of GUI interface softwares. These include objected oriented and procedure oriented technologies as well database languages (DBMS, SQL) in addition to analytics tools.

g) IoT Stack Layer 7: Application layer:

This layer utilizes rest of the six layers in order to develop desired application. The typical case studies or applications of IoT are as follows.

- Smart Home
- Smart Parking System based on zigbee, LoRaWAN and other wireless technologies.
- Smart Energy System, Refer Smart Grid Architecture.
- Smart City
- Smart lighting system based on zigbee standard
- Smart Retail
- Smart Agriculture farming
- Smart Waste Management

1.8.1. The 5 Layers of IoT Technology Stack

The IoT stack consists of the following layers:

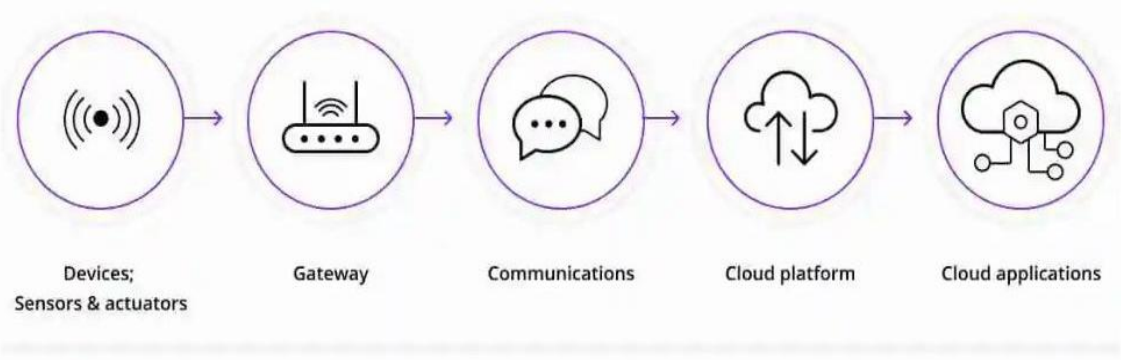


Figure1.17 : 5 layer-IoT stack

- **devices** that include hardware, hardware software,
- sensors and actuators,

- ❖ gateways and communication technologies,

IoT platform that comprises a cloud platform and application software

i) IoT devices: The first layer of the IoT stack is the device hardware that acts as the interface between the physical object and the server (cloud or on-premise).

- ✓ **Sensors and actuators**
- ✓ **Raspberry Pi**
- ✓ **Arduino**

Layer 1: Device Hardware

The first layer of the IoT stack is the device hardware that acts as the interface between the physical object and the server (cloud or on-premise).

Collecting of the data from individual or multiple sensors requires the appropriate type of processor and suitably rated electrical interfaces. This can be as simple as a microcontroller or as complex as an industrial computer.

The device hardware runs the device software and may also include additional sensors, actuators and other input/output (I/O) components.

Sensors measure physical parameters, such as temperature, humidity, pressure or vibration. Actuators can be used to control devices, such as motors, pumps, heaters or lights.

Other I/O components may include buttons, displays or speakers.

If the object or sensors that are to become a 'connected device' lack the suitable processing requirements, then a Single Board Computer (SBC) combined with an appropriate interface cape may be installed as an embedded device.

Layer 1: Device software

- ✓ The second layer of the IoT stack is the device software. This code runs on the device's processor and controls its functionality.

Device software typically consists of an operating system (OS) and application software.

- ✓ The OS configures and manages the device's hardware to provide a platform for the application software to run on.

Common open source and licensed embedded operating systems include Bare metal RTOS, Free RTOS, VxWorks, QNX and Embedded Linux.

- ✓ Application software uses the OS's interfaces to give the device its specific functionality. For

example, an application might be used to control a motor, read sensor data and transmit live or periodic status reports over a wired or wireless connection.

- ✓ In the IoT world the use of open-source software offers immense flexibility and adaptability as business needs and technologies evolve.

ii) Layer2: IoT gateway

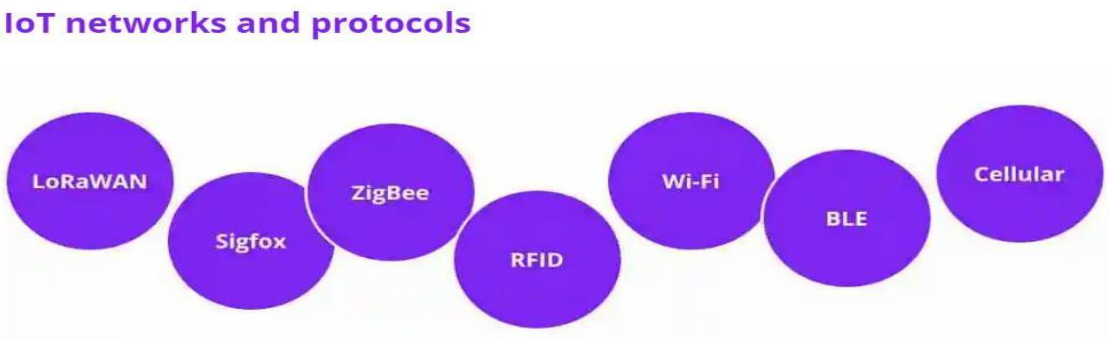


Figure:1.18 Different IoT Networks

An IoT gateway acts as a bridge connecting local Internet of Things (IoT) devices to the cloud, translating various communication protocols and securely transferring data for analysis and remote management. It functions as a central hub, collecting data from numerous sensors and smart devices using short-range technologies like Bluetooth, Zigbee, or Wi-Fi, and then transmitting it over long-range connections such as Ethernet, Wi-Fi, cellular, or satellite to a cloud platform.

In addition to communication, IoT gateways can also provide fog-computing capabilities, processing data locally before sending it to the cloud to enable faster decision-making and reduce latency.

iii) Layer 3: Communications

The third layer of the IoT stack is communications. This is how devices connect to the internet and transfer data.

There are a variety of different technologies that can be used for IoT communications:

- Wired Network
- Wi-Fi
- Bluetooth
- Zigbee/Thread
- 2G/3G/4G/5G cellular networks
- LTE-M/NB-IoT cellular licensed LPWANs (Low Power Wide Arean Networks)

- LoRaWAN, Sigfox (unlicensed LPWANs)

When there is a requirement for devices to communicate with each other, they need to be able to speak the same language. This is where communication protocols come in and come in many different standards published for reference when implementing a new product or service.

Protocols are like rules for how data should be formatted and transmitted. Standard protocols used in the IoT include MQTTS, HTTPS and CoAP.

iv) Layer 4: Cloud Platform

The fourth layer of the IoT stack is the cloud platform. This is the hardware and software services located in internet-based data centers. It is where the data from IoT devices is recorded, captured, processed, analyzed and stored.

Cloud computing in IoT offers a centralized service containing computing resources and databases that can be accessed whenever from wherever.

Unlike on-premise alternatives, the cloud offers ease of scalability in terms of infrastructure capacity and the of device data.

Cloud platforms usually provide tools and services that make it easier to develop and deploy IoT applications.

With IoT services it is advised to compare the Total Cost of Ownership (TCO) when it comes to hosting and running the IoT services in the cloud:

- Bare Metal Servers
- Amazon Web Services (AWS)
- Microsoft Azure

Google Cloud Platform (GCP)

v) Layer 5: Cloud Applications

The fifth and final layer of the IoT stack is cloud applications. These applications run on top of the cloud platform and provide the end-user with a way to interact with the IoT system.

Cloud applications include the user interfaces of dashboards, reporting suites, analytical and artificial intelligence modules. They can be used to view data from sensors and control actuators embedded in static or mobile machinery, plant or fleets of vehicles.

1.9. Sensors in Internet of Things (IoT)

1.9.1 Sensor:

Sensors are used for sensing things and devices etc.

A device that provides a usable output in response to a specified measurement. The sensor attains a physical parameter and converts it into a signal suitable for processing (e.g. electrical, mechanical, optical) the characteristics of any device or material to detect the presence of a particular physical quantity.

The output of the sensor is a signal which is converted to a human-readable form like changes in characteristics, changes in resistance, capacitance, impedance, etc.

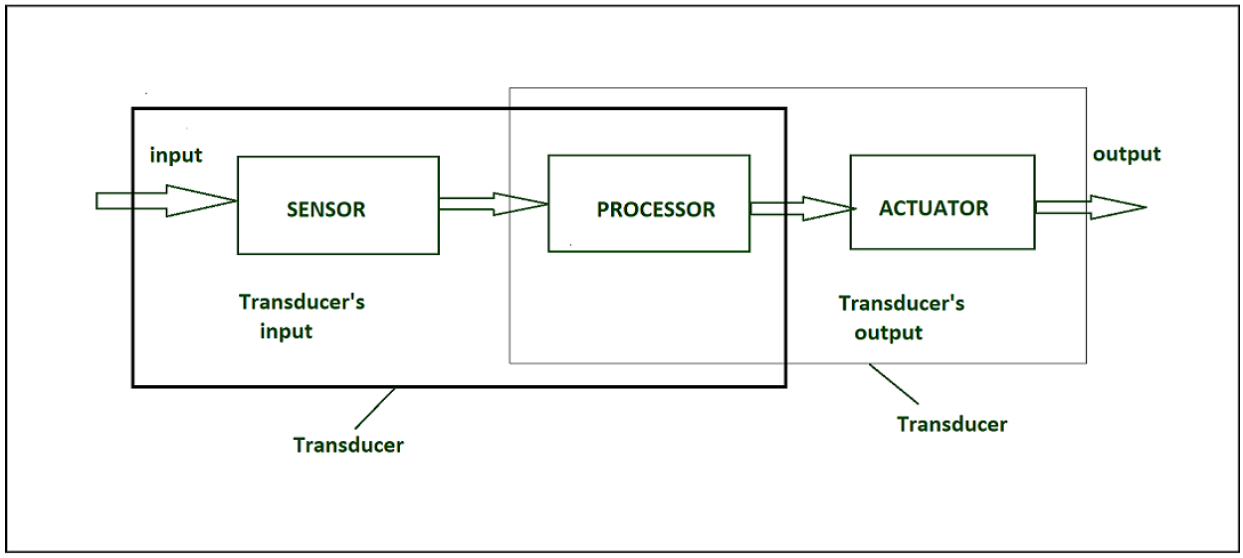


Figure: 1.19: Transducer and Sensor

Transducer:

- A transducer converts a signal from one physical structure to another.
- It converts one type of energy into another type.
- It might be used as actuator in various systems.

1.9.2. Characteristics of Sensors:

1. Static
2. Dynamic

1. Static characteristics:

It is about how the output of a sensor changes in response to an input change after steady state condition.

- **Accuracy:** Accuracy is the capability of measuring instruments to give a result close to the true value of the measured quantity. It measures errors. It is measured by absolute and relative errors.

  Absolute error = Measured value – True value

  Relative error = Measured value/True value
- **Range:** Gives the highest and the lowest value of the physical quantity within which the sensor can actually sense. Beyond these values, there is no sense or no kind of response.
- **Resolution:** Resolution is an important specification for selection of sensors. The higher the resolution, better the precision. When the accretion is zero to, it is called the threshold. Provide the smallest changes in the input that a sensor is able to sense.
- **Precision:** It is the capacity of a measuring instrument to give the same reading when repetitively measuring the same quantity under the same prescribed conditions. It implies agreement between successive readings, NOT closeness to the true value. It is related to the variance of a set of measurements. It is a necessary but not sufficient condition for accuracy.
- **Sensitivity:** Sensitivity indicates the ratio of incremental change in the response of the system with respect to incremental change in input parameters. It can be found from the slope of the output characteristics curve of a sensor. It is the smallest amount of difference in quantity that will change the instrument's reading.
- **Linearity:** The deviation of the sensor value curve from a particularly straight line. Linearity is determined by the calibration curve. The static calibration curve plots the output amplitude versus the input amplitude under static conditions. A curve's slope resemblance to a straight line describes linearity.
- **Drift:** The difference in the measurement of the sensor from a specific reading when kept at that value for a long period of time.
- **Repeatability:** The deviation between measurements in a sequence under the same conditions. The measurements have to be made under a short enough time duration so as not to allow significant long-term drift.

1.9.3. Sensor classifications

Sensor Classified based on

i. **Power Requirements:**

- **Active Sensor –** Independently sense the input. Example- Radar, sounder and laser altimeter sensors.
- **Passive Sensor –**Can not independently sense the input.

Ex- Accelerometer, soil moisture, water level and temperature sensors.

ii. **Sensor output:**

- **Analog Sensor –**The response or output of the sensor is some continuous function of its input parameter.

Ex- Temperature sensor, LDR, analog pressure sensor and analog hall effect.

- **Digital sensor –**Response in binary nature. Design to overcome the disadvantages of analog sensors. Along with the analog sensor, it also comprises extra electronics for bit conversion.

Ex – Passive infrared (PIR) sensor and digital temperature sensor

iii. **Property To be measured**

- **Scalar-** Detects the input parameter only based on its magnitude. The sensor is a function of magnitude of some input parameter. Not affected by the direction of input parameters.

  Example – temperature, gas, strain, colour and smoke sensor.
- **Vector sensor –**The response of the sensor depends on the magnitude of the direction and orientation of input parameter.

  Example – Accelerometer, gyroscope, magnetic field and motion detector sensors.

1.9.4. Sensing Types

Sensing can be divided into four different categories based on the nature of the environment being sensed and the sensors being used;

1. Scalar sensing
2. Multimedia sensing or Vector Sensing

3. Hybrid sensing
4. Virtual sensing

1. Scalar Sensing

Scalar sensing involves the collection of single-value data points from the environment. This type of sensing typically uses traditional sensors that measure a specific physical quantity and output a numerical value.

- **Characteristics:** Focuses on single, quantifiable measurements like temperature, pressure, humidity, light intensity, etc.
- **Sensors Used:** Thermometers, barometers, hygrometers, simple light sensors, etc..
- **Examples:** A temperature sensor reporting the ambient temperature, a pressure sensor measuring tire pressure in a vehicle, or a light sensor detecting the brightness level in a room for automatic lighting control.

2. Multimedia Sensing

Multimedia sensing involves the collection of rich, multi-dimensional data, such as images, videos, or audio. This type of sensing aims to capture complex environmental information that cannot be represented by a single numerical value.

- **Characteristics:** Deals with large volumes of data that often require processing and interpretation to extract meaningful information.
- **Sensors Used:** Cameras (for still images and video), microphones, and other sensors capable of capturing visual or auditory data.
- **Examples:** A security camera recording video surveillance, a microphone capturing sound for environmental monitoring, or a drone equipped with a camera for aerial mapping.

3. Hybrid Sensing

Hybrid sensing combines different types of sensing, typically integrating both scalar and multimedia sensing capabilities. This approach provides a more comprehensive understanding of the environment by correlating various data streams.

- **Characteristics:** Integrates diverse sensor types to gather a more complete picture of the sensed environment. Data fusion techniques are often employed to combine and analyze the

different data streams.

- **Sensors Used:** A combination of scalar sensors (e.g., temperature, pressure) and multimedia sensors (e.g., cameras, microphones) within a single system or network.
- **Examples:** A smart home system that uses temperature sensors and motion detectors along with security cameras and voice control, or an industrial monitoring system that combines vibration sensors with thermal imaging to detect equipment malfunctions.

4. Virtual Sensing

Virtual sensing, also known as soft sensing or inferred sensing, involves deriving information about a parameter that is not directly measured by a physical sensor. Instead, it uses data from other physical sensors and computational models to estimate or predict the value of the unmeasured parameter.

- **Characteristics:** Relies on data processing and modeling rather than direct physical measurement for a specific parameter. Useful when direct sensing is impractical, too costly, or impossible.
- **Methods Used:** Algorithms, mathematical models, and data fusion techniques applied to data from other physical sensors.
- **Examples:** Estimating the concentration of a certain chemical in a reactor based on temperature and pressure readings and a chemical process model, or predicting road conditions (e.g., ice) based on external temperature and humidity measurements without a direct ice sensor

1.9.5. Types of sensors

- **Electrical sensor:** Electrical proximity sensors may be contact or non-contact.
  Simple contact sensors operate by making the sensor and the component complete an electrical circuit.
  Non- contact electrical proximity sensors rely on the electrical principles of either induction for detecting metals or capacitance for detecting non-metals as well.
- **Light sensor:** Light sensor is also known as photo sensors and one of the important sensors. Light dependent resistor or LDR is a simple light sensor available today.

The property of LDR is that its resistance is inversely proportional to the intensity of the ambient light i.e when the intensity of light increases, its resistance decreases and vise versa.

- **Touch sensor:** Detection of something like a touch of finger or a stylus is known as touch sensor. Its name suggests that detection of something.

  They are classified into two types:

  1. Resistive type
  2. Capacitive type

  Today almost all modern touch sensors are of capacitive types. Because they are more accurate and have better signal to noise ratio.

- **Range sensor:** Range sensing concerns detecting how near or far a component is from the sensing position, although they can also be used as proximity sensors.

  Distance or range sensors use non-contact analog techniques. Short range sensing, between a few millimetres and a few hundred millimetres is carried out using electrical capacitance, inductance and magnetic technique.

  Longer range sensing is carried out using transmitted energy waves of various types eg radio waves, sound waves and lasers.

- **Mechanical sensor:** Any suitable mechanical / electrical switch may be adopted but because a certain amount of force is required to operate a mechanical switch it is common to use micro-switches.

- **Pneumatic sensor:** These proximity sensors operate by breaking or disturbing an air flow.

  The pneumatic proximity sensor is an example of a contact type sensor. These cannot be used where light components may be blown away.

- **Optical sensor:** In their simplest form, optical proximity sensors operate by breaking a light beam which falls onto a light sensitive device such as a photocell. These are examples of non-contact sensors. Care must be exercised with the lighting environment of these sensors for example optical sensors can be blinded by flashes from arc welding processes, airborne dust and smoke clouds may impede light transmission etc.

- **Speed Sensor:** Sensor used for detecting the speed of any object or vehicle which is in motion is known as speed sensor.

  For example – Wind Speed Sensors, Speedometer, UDAR (Ultra Deep Azimuthal Resistivity), Ground Speed Radar.

- **Temperature Sensor:** Devices which monitor and tracks the temperature and gives temperature's measurement as an electrical signal are termed as temperature sensors. These electrical signals will be in the form of voltage and is directly proportional to the temperature measurement.
- **PIR Sensor:** PIR stands for passive infrared sensor and it is an electronic sensor that is used for the tracking and measurement of infrared (IR) light radiating from objects in its field of view and is also known as Pyroelectric sensor. It is mainly used for detecting human motion and movement detection.
- **Ultrasonic Sensor:** The principle of ultrasonic sensor is similar to the working principle of SONAR or RADAR in which the interpretation of echoes from radio or sound waves to evaluate the attributes of a target by generating the high frequency sound waves.

1.10. Actuators

An actuator is a mechanical device that converts a source of energy into a controlled physical motion. In simple terms, an actuator is the "mover" of a machine—it takes a signal from a control system and executes a physical action, like opening, closing, pushing, or pulling.

1.10.1. Actuator Characteristics

A set of four characteristics can define all the actuators:

1. Weight
2. Power Rating
3. Torque to Weight Ratio
4. Stiffness and Compliance

i. **Weight:** The physical mass of the actuator, which can limit its application scope. Heavy actuators are typically used in stationary industrial settings, while lightweight ones are essential for mobile systems like robotics, drones, or vehicles.

ii. **Power Rating:** This refers to the power an actuator can consume or deliver, affecting its force and speed capabilities. High-power-rated actuators generally offer faster speeds but might require more significant power consumption.

iii. **Torque to Weight Ratio:** This ratio indicates an actuator's performance, particularly its ability to produce high force or speed relative to its own weight. A higher torque-to- weight ratio signifies better performance and can be critical in applications where both power and lightweight design are needed, like robotics or aerospace.

iv. Stiffness and Compliance:

a. **Stiffness** refers to an actuator's resistance to deformation under load; a high-stiffness actuator maintains its position rigidly.
b. **Compliance** is the ability to deform or adapt under load, often incorporated for safety in human-robot interaction or energy efficiency through elastic elements. These characteristics dictate an actuator's ability to safely and efficiently interact with its environment or handle varying loads.

1.10.2. Classification of actuators

Actuators in the Internet of Things (IoT) can be classified based on two primary criteria: the type of energy source they utilize and the type of motion they generate. This classification helps in understanding their functionality and suitability for different IoT applications, from smart homes to industrial automation.

a. Classification by Energy Source:

1. **Electric Actuators:**

   Mechanism: Convert electrical energy into mechanical motion, often using motors (DC, AC, servo, stepper) or solenoids.

   Advantages: Precise control, programmability, low noise, and relatively clean operation.

   Applications in IoT: Widely used in smart home devices (e.g., smart locks, motorized blinds, smart lighting relays), robotics, and electric vehicles for controlled movements.

2. **Hydraulic Actuators:**

   Mechanism: Use pressurized hydraulic fluid to generate powerful force and motion, typically involving cylinders or fluid motors.

   Advantages: High force and power density, mechanical stiffness.

Applications in IoT: Primarily found in Industrial IoT (IIoT) for heavy-duty machinery like construction equipment, manufacturing robots, and automated presses where significant force is required.

3. **Pneumatic Actuators:**

   Mechanism: Convert compressed air or gas energy into mechanical motion. **Advantages:** Fast response times, durability in harsh environments, suitable for high- speed operations and explosion-proof areas.

   Applications in IoT: Used in industrial automation (e.g., controlling robotic grippers, pneumatic valves), and some smart irrigation systems.

4. **Thermal Actuators:**

   Mechanism: Utilize temperature-sensitive materials (like shape memory alloys) that change shape or size with temperature variations to produce motion.

   Advantages: Compact size, high power density, and potential for integration without separate power sources in some cases.

   Applications in IoT: Smart HVAC systems (e.g., thermostats, thermal bypass valves), and some micro-electromechanical systems (MEMS).

5. **Magnetic Actuators:**

   Mechanism: Generate force and motion using magnetic fields, including electromagnets or permanent magnets.

   Advantages: Non-contact operation, precise control in certain applications. **Applications in IoT:** Smart locks, specialized valves, and precision positioning systems in various smart devices.

b. Classification by Type of Motion:

1. **Linear Actuators:**

   Motion: Produce movement along a straight line (pushing, pulling, lifting).

   Examples in IoT: Automatic doors, smart window openers, adjustable standing desks, and conveyor belt systems in smart factories.

2. **Rotary Actuators:**

   Motion: Generate rotational or angular movement.

   Examples in IoT: Electric motors in robotic arms, smart valves controlling fluid flow, and motorized camera gimbals in surveillance systems.

3. **Oscillating Actuators:**

   Motion: Produce a repetitive back-and-forth or angular movement within a limited range.

   Examples in IoT: Vibrating components for haptic feedback in wearables or certain industrial mixing applications

1.10.3. Actuating Types:

In the context of the Internet of Things (IoT), actuating types play a critical role in enabling physical actions based on data and decisions made within the IoT ecosystem. While the fundamental principles of actuation remain the same

1. **Electromechanical Actuators (Most Common in IoT):**

   Motor-Driven (e.g., in Smart Homes/Buildings): Electric motors combined with mechanisms like lead screws or gears are extensively used in smart blinds, automated door locks, smart valves (for water or gas), robotic vacuum cleaners, and automated climate control systems where precise positioning or controlled movement is required based on sensor input or user commands via an IoT platform.

   Solenoids (e.g., in Smart Locks/Access Control): Solenoids are frequently used in IoT- enabled smart locks for quick, on/off locking/unlocking actions, remotely controlled via mobile apps or integrated into smart home security systems. They are also found in smart irrigation valves.

   Piezoelectric Actuators (e.g., in Micro-IoT Devices): Due to their precision and small size, piezo actuators are used in specialized IoT applications requiring very fine control, such as in micro-fluidic devices for smart healthcare, precision dosing in smart agriculture, or haptic feedback in wearables.

   Electrical Actuators

   - **Principle:** Convert electrical energy into mechanical motion using electric motors (AC or DC).
   - **Components:** Include an electric motor, gearing (if needed for torque/speed conversion), and a control system.
   - **Characteristics:** Provide precise control over position, speed, and force, often quieter than fluid-powered systems, and integrate easily with electronic control systems.
   - **Applications:** Found in a vast range of applications from industrial robots and

automated assembly lines to smart home devices and electric vehicles.

Mechanical Actuators

- **Principle:** Convert rotary motion into linear motion or vice-versa using physical mechanisms like gears, screws, levers, or cams.
- **Components:** May include motors (electric), gears, lead screws or ball screws, and linkages or levers.
- **Characteristics:** Known for simplicity, durability, and cost-effectiveness in certain applications, often used where manual operation is an option or precision is less critical.
- **Applications:** Used in a variety of simple machines, manual valve systems, mechanical locks, and systems where cost and robustness are key considerations.

2. Fluid Power Actuators (Less Common but Present in Industrial IoT): Hydraulic/Pneumatic Systems (e.g., in Industrial IoT/Smart Factories): While less common in consumer IoT, these are vital in Industrial IoT (IIoT). Pneumatic and hydraulic actuators are controlled by IoT systems for automated manufacturing lines, robotic arms, and heavy machinery, enabling remote monitoring, predictive maintenance, and optimized operation based on real-time data from sensors. Smart valves controlled by IoT gateways are key here.

Hydraulic Actuators

- **Principle:** Operate on the principle of fluid pressure, converting hydraulic fluid pressure into mechanical motion, usually linear or rotary.
- **Components:** Typically consist of a cylinder, piston, and a hydraulic power unit (pump, reservoir, valves).
- **Characteristics:** Known for generating high forces and torques, robust construction, and ability to hold position under load.
- **Applications:** Commonly used in heavy machinery like excavators, industrial presses, and manufacturing equipment requiring high force and precision.

Pneumatic Actuators

- **Principle:** Utilize compressed air or gas to generate mechanical motion, similar to hydraulic actuators but using a compressible medium.
- **Components:** Feature cylinders, pistons or diaphragms, and an air compressor system.

- **Characteristics:** Offer fast response times, cleanliness (no fluid leaks), and suitability for hazardous environments as they don't produce sparks.
- **Applications:** Widely used in industrial automation for tasks like valve control, material handling, and robotics where quick, repetitive movements are needed.

3. **Specialized Actuators:**

Smart Materials & Thermal Actuators (e.g., in Wearables/Smart Fabrics): As IoT expands into wearables and smart textiles, thermal or shape memory alloy (SMA) actuators can be used for dynamic adjustments in clothing (e.g., adaptive ventilation) or in medical devices for controlled drug release.

Magnetic Actuators: Used in specialized IoT applications requiring non-contact force application or precise, small-scale movement, such as in certain smart valves or micro-robotics.

Thermal / Magnetic Actuators

- **Principle:**
    - Thermal: Utilize the expansion or contraction of materials in response to temperature changes to generate motion.
    - Magnetic: Rely on magnetic fields to induce force or motion, often involving solenoids or electromagnetic coils.
- **Components:** Vary widely depending on the specific type; thermal actuators may involve bimetallic strips or shape-memory materials, while magnetic actuators use coils and magnetic cores.
- **Characteristics:** Often compact, lightweight, and can be used for precise, small-scale movements.
- **Applications:** Common in micro-electromechanical systems (MEMS), sensors, and specialized applications requiring temperature-dependent or magnetic field-induced actuation, such as in medical devices or fluid control.

4. **Other actuators Soft Actuators**

- **Principle:** Made from compliant, flexible materials (e.g., polymers, elastomers) that

deform in response to various stimuli like fluid pressure, electricity, light, or temperature.

- **Components:** Typically involve flexible structures, embedded channels for fluid, or electroactive materials, often without rigid moving parts.
- **Characteristics:** Offer high adaptability to complex shapes and environments, safe interaction with delicate objects or biological tissues, and potential for untethered operation.
- **Applications:** Emerging field with applications in soft robotics, wearable devices, biomedical tools, and human-machine interfaces.

Shape Memory Polymers (SMPs)

- **Principle:** A type of smart material that can be programmed to remember and recover a specific "permanent" shape when triggered by an external stimulus (most commonly heat, but also light, electricity, or other factors).
- **Components:** Consist of polymeric networks with reversible crosslinks or transitions that enable shape fixation and recovery.
- **Characteristics:** Exhibit high deformability, tailored activation temperatures, and can be designed for one-way or multi-shape memory effects.
- **Applications:** Potential uses in biomedical devices (e.g., stents, sutures), smart textiles, deployable structures, and self-healing materials.

1.10.4. List of actuators used in IoT

- **Electrical Actuators**
  - DC Motor
  - Stepper Motor
  - Servo Motor
  - Solenoid
  - Relay
- **Mechanical Actuators**
  - Cam and Follower
  - Rack and Pinion
  - Chain Drive

- Gear Mechanism

- **Pneumatic Actuators (air powered)**
  - Pneumatic Cylinder
  - Pneumatic Motor
  - Air Bellows Actuator
- **Hydraulic Actuators (fluid powered)**
  - Hydraulic Cylinder
  - Hydraulic Motor
  - Hydraulic Pump Actuator
- **Thermal / Magnetic Actuators**
  - Bimetallic Strip (in thermostat, iron)
  - Shape Memory Alloy Actuator
  - Electromagnet